Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/15/2024
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Project Title:	Understanding the Frequency Synchronization Physics in PK-4 Experiment		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
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Zip Code:	20375-0001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crabtree, Chris Ph.D. (Naval Research Laboratory)		
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Task Description:	The objective of this proposal is to provide theoretical support for a separate PK-4 (Plasma Krystall-4) experimental project, “Nonlinear Waves in Dusty Plasmas.” The experimental effort, led by Principal Investigator (PI) Goree, is an investigation of nonlinear synchronization, with the advance that it will study not point oscillators, but waves that propagate a distance of many wavelengths. The literature is lacking theory suitable for comparison to the experiment, perhaps because the experiments at this point are still very new. In particular, theory is needed for the case, beyond the point oscillator description that is most common, for synchronization in the realm of propagating waves. Beyond just supporting the PK-4 experiment, the theoretical advances we propose can have a broad interdisciplinary impact. Since synchronization, frequency entrainment, and frequency pulling are nonlinear phenomena that impact a variety of physical, chemical, and biological systems, the proposed investigation will be relevant to a broader area of contemporary interest and enhance the science foundation for nonlinear phenomena in plasma physics. A plasma system normally constitutes a highly complex medium that can display a rich variety of nonlinear phenomena arising out of a host of intrinsic oscillations and other non-stationary states. A clear identification of phenomena like synchronization or frequency entrainment in such a medium often poses serious experimental challenges. Many past studies on such systems have therefore relied on driving a single plasma system with an external frequency source. There have been very few studies devoted to the investigation of mutual synchronization between two plasma systems except where this phenomenon was studied between two double plasma devices and where it was investigated in two electrical discharges. There is no theoretical framework to account for these phenomena in a realistic dispersive plasma system that includes natural growth and dissipation of waves. Therefore, the primary focus of this proposal is to develop such a framework and apply it to understand the dedicated PK-4 dusty plasma experiment on frequency synchronization. The validated model will then be useful for general application to other experiments. Some of the science questions that we will address in our proposed theoretical investigation are the following: • What are the parametric domains in the state space of the external driver frequency and amplitude that lead to frequency entrainment, frequency pulling, and synchronization of the dust acoustic waves in the PK-4 plasma? • How do dispersion and nonlinearity of the dusty plasma medium influence the nature of the synchronization process? • Can the non-harmonic synchronized state, as observed in ground-based experiments, be predicted by a continuum model of the dusty plasma's constituents? • What determines the threshold amplitude for synchronization, and how is the threshold affected by dust-dust correlations (i.e., the strength of the Coulomb parameter)? • How does viscous dissipation in the medium affect synchronization and influence the nature of the synchronized excitation? • Can one observe phase-flip transitions in the synchronization, i.e., a sudden transition between in-phase synchronization to pi out-of-phase synchronization? • How does a broad-spectrum driver (e.g., signals from another plasma source) alter the synchronization scenario?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Beyond just supporting the PK-4 experiment, the theoretical advances in this project will have a broad interdisciplinary impact. Since synchronization, frequency entrainment, and frequency pulling are nonlinear phenomena that impact a variety of physical, chemical, and biological systems, the investigation is relevant to a broader area of contemporary interest.
Task Progress:	This basic research is a theoretical study of dusty plasma physics in close conjunction with an experimental group. The experimental effort, led by Principal Investigator (PI) J. Goree, University of Iowa, is an investigation of nonlinear synchronization of frequencies of natural dust acoustic oscillations in the Plasma Crystal (PK-4) module with an externally imposed signal. The PK-4 experiment was conducted in the zero-gravity environment of the International Space Station (ISS). Ground-based laboratory experiments were carried out in the University of Iowa to prepare for the microgravity experiments in the PK-4 module. We developed nonlinear models relevant to the PK-4 experimental project, “Nonlinear Waves in Dusty Plasmas.” The novelty of this theoretical investigation was that a more realistic case, including the effects of waves in a plasma that propagate a distance of many wavelengths in the plasma, was considered instead of the usual point oscillators. Our research began with a review of the related materials pertinent to the PK-4 experiment. We fit the experimental data with an exact nonlinear solution of the KdV equation, namely the conodal wave solution, and obtained an excellent agreement between the experimental observations and the model solutions over a large range of the background pressure. The success of the nonlinear equation for conodal solutions depends on the right approximation of the physics of the collective effects. To ascertain that the physics of the dust acoustic oscillation due to relative streaming of ions and electrons with respect to dust was properly captured, we first derived the general kinetic dispersion relation relevant to the experimental system. With accurate knowledge of the instability conditions, we modeled the nonlinear evolution of the excited dust acoustic waves due to ion streaming. Based on the good agreement between experimental observations of spontaneous dust acoustic waves and the exact conoidal wave solution of the KdV model equation, we proceeded to explore the feasibility of applying this model to study the synchronization phenomenon. The model was extended by adding a time varying source as a driving term and numerical solutions of the equation obtained over a range of the driving frequency. However, although our preliminary numerical investigations revealed a host of diverse and rich nonlinear results, it failed to observe any synchronization phenomenon. This was puzzling initially, but became clear under further scrutiny of the KdV model and on examining the necessary mathematical conditions for the onset of synchronization. A basic requirement for synchronization is the existence of limit cycles in the solution space of the non-driven model towards which the source term can drive the system. Limit cycles can exist only in the presence of dissipation (energy sink) and a driving term (energy pump) in the system. Physically, this meant that we needed to consider the frictional losses in the system (e.g., due to collisions and/or viscosity). The simple KdV model did not have such a term. Accordingly, we derived a new nonlinear model that considered viscosity contributions and arrived at a KdV-Burger equation. Numerical solution of this equation was successful in simulating the synchronization phenomenon. The generalized study was based on the forced Korteweg-de Vries-Burgers (fKdV-B) equation, which is a generalization

of the forced Korteweg-de Vries equation (fKdV), by incorporating dissipative effects arising from viscosity in the medium. This equation provides a proper theoretical framework and a better physical model, compared to the Van der Pol oscillator model for studying the dynamics of nonlinear dust acoustic waves, by properly accounting for nonlinear, dispersive, and dissipative influences on the waves. Using the model, we have successfully demonstrated harmonic (1:1) and super-harmonic (1:2) synchronization states of dust acoustic waves (DAWs) for the experimental values reported in the laboratory experiment of Ruhunusiri et al. [Ed. Note: See References below.] In particular, comparison of our theoretical Arnold tongue diagram with their experimental one shows the following common features. As in the experimental Arnold tongue diagram, we see the existence of amplitude thresholds as well as clear evidence of the branching phenomena. However, there are also important differences. With our model, we have not been able to obtain sub-harmonic synchronization that have been observed in the experiment. Furthermore, our model uses an external driver that closely resembles a nonlinear natural mode of the system; whereas, in the experiment, a purely time varying external sinusoidal driver has been used. However, it is not clear what form this driver takes inside the plasma system and whether it manifests itself as a spatio-temporally varying perturbation. These and other questions, such as the absence of sub-harmonic synchronization in the equation, the neglect of dissipation arising from gas friction on the dust particles, etc., remain to be explored in the future in order to further improve the model.

Keeping in mind the broad interdisciplinary nature of our project, we also looked at the interesting results obtained with the sinusoidally driven KdV equation (i.e., in the absence of dissipation). Using an exact analytic solution of such a model, we found that the model can be usefully deployed to explain nonlinear wave mixing and harmonic generation in a wide variety of nonlinear dispersive systems where the KdV approximation holds. Thus, this provides a new paradigm to explain some of the earlier experimental and numerical results obtained for laser plasma interactions.

REFERENCES

Ruhunusiri WS, Goree J. Synchronization mechanism and Arnold tongues for dust density waves. Physical Review E. 2012 Apr 2;85(4):046401.

Bibliography Type: Description: (Last Updated: 03/22/2023)